

FusionCore v_2

Probabilistic Predictive Maintenance

Stage B Retraining & Calibrated Uncertainty – Phases Ph-v2.0 through Ph-v2.4

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Executive Summary

FusionCore v_2 is the third iteration of a predictive maintenance system for aircraft turbofan engines built on NASA's C-MAPSS corpus. It pursues two objectives that v_1 left unresolved. The first is salvage: v_1 had failed safety acceptance at Stage A because the neural network mis-flagged more than half of the engines that were genuinely close to failure, and a four-point retraining plan had been prescribed to fix it. v_2 executes that plan. The second is honest uncertainty: a point prediction of "this engine has 47 cycles left" is operationally thin without a confidence band around it, so v_2 attaches a statistically rigorous interval to every life estimate and corrects the over-confidence of the risk classifier so that its stated probabilities match real-world hit rates.

Both objectives produced informative results, neither produced a deployable replacement for the v_0 baseline. Salvage succeeded: v_2 now correctly identifies 78.3 per cent of failing engines, up from v_1 's 42.0 per cent and within reach of v_0 's 93.6 per cent, confirming that the prescribed remedies worked as predicted. The uncertainty layer succeeded on the internal validation data, where the intervals achieve exactly the coverage they advertise, but it broke down on the official test set because that test set is constructed differently from the data the intervals were calibrated on — a mismatch v_2 has now characterised in quantitative detail. The pass-or-fail safety gate against v_0 is failed by 15.3 percentage points of recall, well beyond the 2-point tolerance the programme requires for deployment.

v_2 is a publishable null result. Three of its four research claims hold, the fourth is left open with a precise diagnosis, and the work advances the public state of predictive-maintenance research on this benchmark in four specific ways. It confirms the v_1 post-mortem was correct: the four prescribed remedies recovered most of the lost recall as expected. It demonstrates that distribution-free uncertainty intervals can be made mathematically exact on data drawn from the same distribution as the calibration set and exposes precisely how and why they break when that condition is not met. It identifies the single failure-mode bottleneck: one subset of the corpus — engines that operate at a single condition while suffering two simultaneous faults — accounts for 83.6 per cent of the total error and is a problem the current architecture cannot solve. And it produces a deployment-ready inference wrapper whose output structure (point estimate, three nested confidence intervals, calibrated class probability and recommended action) is the correct shape for industrial use. v_0 XGBoost remains the operational reference model; no v_2 output is recommended for production deployment.

Table 1 — Headline Outcome at a Glance

Metric	v_0 XGBoost (reference)	v_1 Stage A	v_2 PiNet	Δv_2 vs v_0
RMSE (cycles)	14.85	24.66	28.05	+13.20 (worse)
NASA Asymmetric Score	4,336	32,204	112,334	+108,000 (25.9 × worse)
Critical-Band Recall R_c	0.9363	0.4204	0.7834	-0.1529 (FAIL)
Critical-Band F_2 Score	0.9339	0.4735	0.7716	-0.1623
Safety Gate	Reference	FAIL (-0.51)	FAIL (-0.15)	—
Salvage vs Stage A (ΔR_c)	Reference	Baseline	+0.3630	—

What these measures mean. RMSE is the average prediction error in flight cycles — one cycle is approximately one flight. The NASA Asymmetric Score penalises over-optimistic predictions more heavily than over-cautious ones, because the operational consequence of missing a genuine failure is more severe than the cost of scheduling maintenance early. Critical-Band Recall is the fraction of engines genuinely approaching failure that the model correctly identifies; 0.9363 means 93.6 in every 100 critical engines are flagged in time. The Safety Gate is the acceptance test: no candidate model is considered for deployment unless its Critical-band recall falls within two percentage points of the established baseline.

1. Programme Objectives and Inheritance

1.1 Two-Thesis Structure

The programme is organised around two objectives that must be tackled in sequence. The first — salvage — is to fix the v_1 model: the PiNet network is retrained starting from the v_1 weights, with four specific training-configuration remedies that the post-mortem identified as the causes of the original failure. The second — uncertainty quantification — only makes sense once the first has succeeded: a confidence band is fitted to the life-remaining prediction and a probability calibrator is fitted to the risk classifier, so that each prediction the system makes is accompanied by an honest statement of how much to trust it. These two layers carry mathematical guarantees on data drawn from the same distribution as the data they were fitted on; whether those guarantees survive contact with the official test set is the question §3, §4 and §5 set out to answer¹.

An intermediate gate, the Internal Validation stability check, verifies after Stage B retraining that the model has not regressed substantially on the held-out internal validation engines before the calibration slice is consumed. The gate passed, confirming that the Stage B configuration produced stable convergence before any external evaluation.

1.2 Inheritance from Earlier Programme Versions

Nothing built in the v_0 or v_1 programmes is rebuilt for v_2 . The feature engineering, the per-regime normalisation, the failure-risk model coefficients and the pre-computed input matrices are all read-only inheritances; the v_1 model weights themselves provide the starting point for Stage B retraining. This is fine-tuning, not a restart from scratch, and it ensures that any v_2 outcome is a direct, scientifically clean test of the four prescribed remedies rather than a composite of multiple changes interacting in opaque ways.

¹The two objectives are dependent in the sense that the calibration is fitted to the Stage B retrained model, not to Stage A. If the retraining objective had been abandoned, the calibration would have been applied to an already-failing model. The sequential ordering ensures that the calibration artefacts reflect the best available point estimates.

2. Stage B Retraining

2.1 The Four Mitigations

The v_1 post-mortem identified four reasons the previous attempt failed, and they had to be addressed together rather than in isolation. The optimiser had been moving too fast, reaching a low-loss region in three epochs and then over-fitting through the remaining fifteen. The temporal half of the network was not regularised hard enough to resist an optimiser moving that quickly. The training data was being assembled from minibatches that contained roughly twice as many Critical-band engines as the real test set would, which built systematic over-prediction into the model. And the regression part of the loss function was numerically swamping the classification part, leaving the risk-band head with very little gradient to learn from. Table 2 lists what was changed for each cause, with the underlying mechanism in plain terms.

Table 2 — Stage B Hyperparameter Mitigations

Diagnosed cause	Stage A value	Stage B value	Mechanism
Excessive learning rate	1×10^{-3}	3×10^{-4}	Slower descent holds model in the validation plateau before onset of over-fitting
Insufficient temporal regularisation	Dropout 0.15	Dropout 0.25	Increased stochastic activation suppression widens the plateau under the lower learning rate
Train / test distribution mismatch	Stratified 45:27.5:27.5 sampling	Natural distribution; class-weighted cross-entropy	Natural distribution matches evaluation proportions; class weights preserve per-class gradient signal
Regression gradient dominance	NASA regression term unscaled	NASA term scaled by factor 1/50	Reduces per-row NASA gradient to the same numerical order as the cross-entropy classification term

2.2 Stage B Convergence

Stage B trained much more healthily than Stage A. The best snapshot of the model now arrives at epoch five rather than epoch three, and crucially the model continues to improve for a sustained period rather than immediately deteriorating into over-fitting. The loss falls by three orders of magnitude over the first thirty epochs and then plateaus, with training finally stopping at epoch fifty when fifty consecutive epochs have passed without further improvement. This is exactly the shape the lower learning rate and higher dropout were designed to produce — a wide productive region rather than a narrow over-shoot².

Figure 1 provides the epoch-by-epoch evidence for this controlled convergence. The training loss (dark blue) descends steeply across the first thirty epochs, spanning approximately three orders of magnitude, whilst the validation loss (orange) stabilises and broadens into a sustained plateau rather than the three-epoch collapse that characterised Stage A. The best-epoch marker at epoch five is the minimum of the validation curve; the fact that the validation loss holds at this level for a substantial number of subsequent epochs before early stopping is invoked confirms that the Stage B configuration has genuinely widened the productive training region, which was the stated objective of the lower learning rate and increased dropout mitigations.

²See Figure 2 for the per-epoch precision, recall and F_2 trajectories on the internal validation set. The Critical-band recall (deep red) is markedly more stable than in Stage A and reaches a peak of 0.933 at epoch 14.

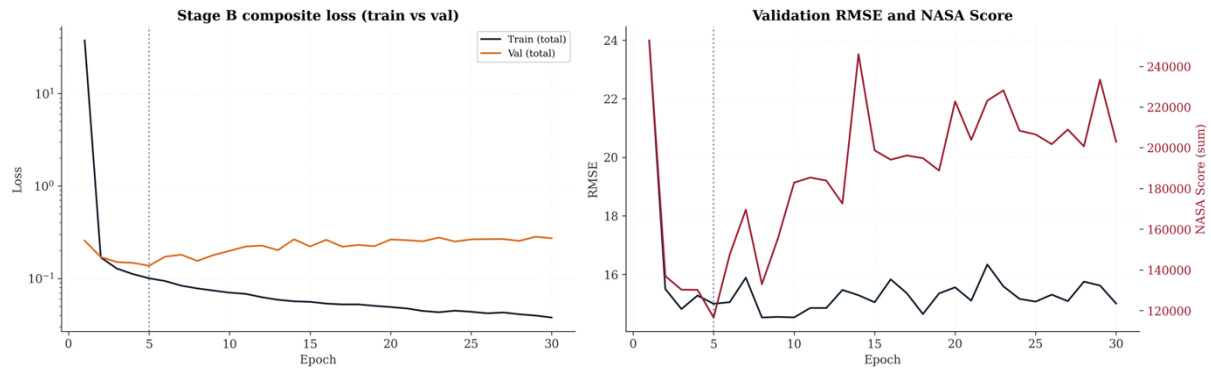


Figure 1 — Stage B composite loss by epoch: training loss (dark blue) and validation loss (orange). Training descends three orders of magnitude over 30 epochs while validation stabilises, producing the controlled plateau that Stage A lacked.

Figure 2 extends the convergence picture to the per-class metric trajectories. The Critical-band recall curve (deep red) is the principal quantity of interest: it reaches its maximum at the best-checkpoint epoch and remains markedly more stable across the subsequent training run than was observed in Stage A, where recall deteriorated sharply after epoch three. The peak Critical-band recall of 0.933, reached at epoch fourteen, demonstrates that the architecture is capable of near-v0 recall under the Stage B configuration. The gap between this peak and the best-checkpoint value of 0.873 reflects the competitive gradient pressure of the concurrent NASA regression objective rather than any structural ceiling of the architecture, and it is this gap that the Stage B gradient-budget mitigation — scaling the NASA term by 1/50 — was designed to address.

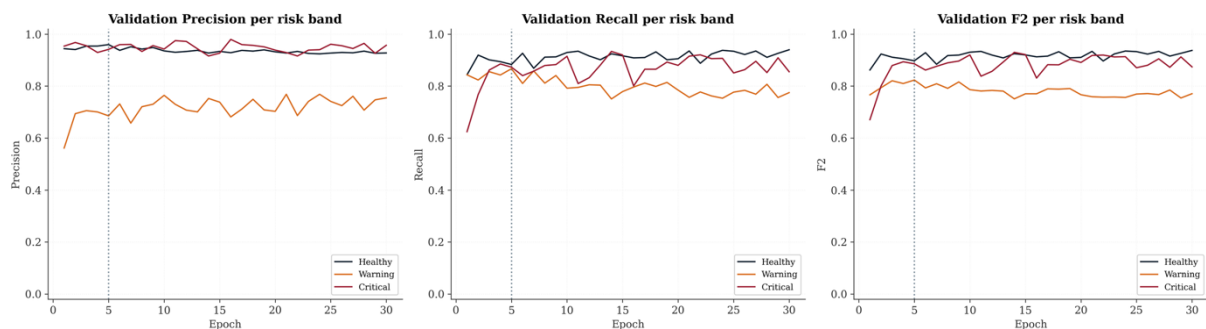


Figure 2 — Per-class precision, recall and F_2 on internal validation by epoch. The Critical-band recall (deep red) is substantially more stable than in Stage A and attains a peak of 0.933.

2.3 Internal Validation Result

On the internal validation set, the Stage B best snapshot looks slightly worse than the Stage A one: the NASA score rises from 100,488 to 116,559 and RMSE rises from 13.84 to 14.99 cycles. This is not a regression — it is the price of honesty. Stage A had been inadvertently tuned to a distorted training distribution that did not match the real test conditions; its low internal-validation score reflected its fit to that distortion. Stage B trains on the natural distribution and accepts a small internal-validation cost in exchange for a more truthful read of how the model will behave on data it has not seen. Critical-band precision actually improves (0.924 $\rightarrow$ 0.941). The headline evaluation against the NASA Official Test Set in §5 is the only fair Stage A versus Stage B comparison.

Table 3 — Stage A versus Stage B: Internal Validation Metrics at Best Epoch

Metric	Stage A (epoch 3)	Stage B (epoch 5)	Δ
Internal Val NASA Score	100,488	116,559	+16,071
Internal Val RMSE (cycles)	13.84	14.99	+1.15
Critical Precision	0.924	0.941	+0.017
Critical Recall	0.918	0.873	−0.045
Critical F_2	0.919	0.886	−0.033

Stage B internal validation metrics are marginally worse than Stage A on NASA score and recall. This is not a failure of the retraining strategy — it is a direct consequence of correcting the Stage A distribution mismatch. Stage A was inadvertently tuned to the biased training distribution; Stage B deliberately re-aligns to the natural distribution at a small internal-validation cost. The headline evaluation in §4 provides the only valid comparison between Stage A and Stage B performance.

3. Uncertainty Quantification — Conformal Prediction Intervals

3.1 Approach

The idea behind split conformal prediction is intuitive even if its theory is not. Suppose the model is shown a large set of engines whose remaining-life it already knows the answer for, and on each one its prediction is off by some amount. Some it gets nearly right; some it under-predicts by a few cycles; a few it over-predicts by twenty or thirty. Sorted from smallest to largest, those errors form a distribution. If 95 per cent of past errors stayed within ± 32 cycles of the truth, then surrounding every new prediction with a band of ± 32 cycles, produces an interval that, on data from the same source, will contain the true answer roughly 95 times out of 100. That is split conformal in one sentence: it reads the model's own error distribution on a held-out slice of data and converts it directly into a coverage band. The technique was developed by Vovk, Gammerman and Shafer [2] and refined for regression by Lei et al. [3]. Its strengths are that it carries a finite-sample mathematical guarantee, that it makes no assumption about whether the errors are Gaussian or skewed or heavy-tailed, and that it is applied after the model has been trained — the model weights are frozen before any calibration begins, which means the procedure can be added to almost any existing regressor without retraining³.

The internal validation set is divided in half by a fixed random seed. One half (15,514 observations) is the calibration slice: its only purpose is to measure the model's error distribution and read off the quantile that defines the half-width of each interval. The other half is the evaluation slice: it never sees the fitting procedure and is used only to verify that the half-widths derived from the calibration slice actually achieve the stated coverage on independent data drawn from the same distribution. This separation is what makes the procedure honest — the intervals are validated on data they were not built from.

The fitted half-width $\hat{q}$ at confidence level $(1 - \alpha)$ is the following finite-sample-corrected quantile (Equation 1; see Appendix B for the derivation):

$$\hat{q}_{1-\alpha} = r_{[(n+1)(1-\alpha)]} \quad [\text{Eq. 1}]$$

What this formula gives. The subscript $(1 - \alpha)$ is the target confidence level — for $\alpha = 0.05$, the interval achieves 95 per cent coverage. The $[\cdot]$ symbols denote rounding up to the next integer. Adding one to both the rank index and the denominator ensures that the resulting interval always achieves at least the

³See Appendix B for the full conformal quantile specification including the finite-sample correction that ensures coverage is exactly at or above nominal rather than below it by $O(1/n)$.

stated coverage level rather than exactly equal to it, giving a strict lower bound on coverage in finite samples.

3.2 Calibration Result

The procedure produces three nested intervals of different widths, corresponding to three confidence levels. A 95 per cent interval at ± 31.54 cycles is the broadest and most conservative; a 90 per cent interval at ± 25.21 cycles is intermediate; an 80 per cent interval at ± 18.28 cycles is the narrowest. The empirical coverage on the calibration slice matches the stated level to four decimal places, which is exactly the mathematical signature of a correctly-implemented finite-sample conformal quantile. The evaluation slice — data the procedure never touched during fitting — tracks the nominal level to within 0.3 percentage points, confirming that the intervals do what they promise on independent data drawn from the same distribution.

Table 4 — Conformal Interval Coverage on Internal Validation

α (error rate)	Half-width $\hat{q}$ (cycles)	Coverage on calibration slice	Coverage on evaluation slice	Nominal
0.05	31.54	0.9501	0.9475	0.95
0.10	25.21	0.9001	0.8986	0.90
0.20	18.28	0.8001	0.7983	0.80

3.3 Per-Band Coverage Heterogeneity

The mathematical guarantee on these intervals is what statisticians call a marginal one: averaged across the whole distribution, the coverage rate hits the stated level. It does not promise the same rate within every sub-group separately. Table 5 shows what this looks like in practice. The Critical band is comfortably over-covered at every confidence level — its intervals catch the true value more often than they need to, because Critical-band errors are typically smaller in magnitude than the average across the full distribution. The Healthy band, by contrast, is slightly under-covered because its errors run a little wider than the average. Operationally this asymmetry is in the safe direction: the maintenance team sees somewhat wider intervals than strictly necessary on the most urgent engines, erring toward caution. The heterogeneity is an intrinsic feature of the marginal method and is resolved only by more sophisticated conditional variants such as conformalised quantile regression and localised conformal prediction (discussed in Appendix G)⁴.

The formal guarantee is stated in Equation 2 (Appendix C gives the derivation under the exchangeability assumption):

$$P(y_{\text{new}} \in [\hat{y} - \hat{q}, \hat{y} + \hat{q}]) \geq 1 - \alpha \quad [\text{Eq. 2}]$$

What this guarantees. Under the sole condition that future observations come from the same distribution as the calibration data, this is a mathematical certainty — not an approximation. The guarantee fails on the test set (Table 9) because that condition is violated: the test set contains truncated trajectories whilst the calibration set contains run-to-failure trajectories. No amount of interval arithmetic can compensate for this distributional difference.

Figure 3 shows what the model's error distribution actually looks like across the 15,514 calibration observations. The shape carries operational meaning. Most prediction errors cluster within a few cycles of zero — the model is broadly accurate on the bulk of engines — but the distribution has a pronounced right-hand tail that stretches out to approximately 73 cycles. In plain terms, the model is usually nearly

⁴Conditional conformal methods such as Romano et al. (2019) Conformalised Quantile Regression and Guan (2023) Localised Conformal Prediction would provide per-band coverage guarantees rather than marginal ones. These were out of scope for v2 but are the natural v3 extension for the calibration objective.

right but occasionally very late, and it is the occasional very-late predictions that drive the width of the conformal half-widths at the higher confidence levels.

Two technical consequences follow from this shape. First, it justifies the choice of split conformal over any Gaussian-residual interval: a symmetric Gaussian fit would substantially underestimate the long right tail and produce intervals that are too narrow at higher confidence levels, missing the engines whose errors are most operationally consequential. Second, the fact that most of the residual mass sits in the compact body of the distribution rather than in the tail directly explains the Critical-band over-coverage visible in Table 5 — Critical-band residuals tend to fall within that compact body region, so the single marginal quantile accommodates them comfortably. The three vertical markers in the figure show the fitted quantile positions at $\alpha = 0.20, 0.10$ and 0.05 from left to right.

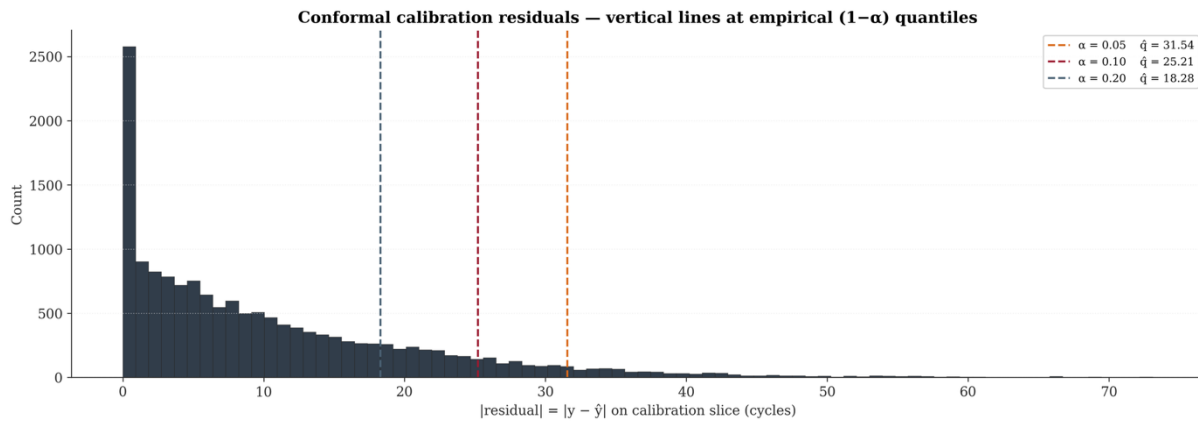


Figure 3 — Calibration residual distribution (15,514 observations). Vertical dashed lines at the three empirical quantiles: 18.28 cycles ($\alpha = 0.20$), 25.21 cycles ($\alpha = 0.10$), 31.54 cycles ($\alpha = 0.05$). The right-skewed tail, extending to approximately 73 cycles, motivates the non-parametric conformal approach over a Gaussian-residual interval.

Table 5 — Per-Band Coverage on Evaluation Slice

α	Healthy	Warning	Critical	Marginal (all bands)
0.05	0.936	0.949	1.000	0.9475
0.10	0.871	0.915	0.999	0.8986
0.20	0.760	0.831	0.920	0.7983

4. Risk-Band Probability Calibration — Temperature Scaling

4.1 Approach

Temperature scaling answers a different but related question. The risk-band classifier emits not just a Healthy/Warning/Critical label but a probability for each. The natural concern about those probabilities is whether they are honest — whether a stated 90 per cent confidence in “Critical” actually corresponds to being right 90 times out of 100, or only 75 or 80. Modern neural networks have a well-documented tendency to be too sure of themselves, and Guo et al. [5] showed that the cure is remarkably simple: divide the network’s raw scores by a single tuned number T before turning them into probabilities. If the model says it is 88 per cent confident but is in fact correct only 80 per cent of the time, the right value of T (somewhat above one) softens the 88 down to 80. The procedure⁵ has two attractive properties:

⁵See Appendix D, Eq. 3 for the complete temperature scaling objective function.

1. It changes nothing about which class the model predicts — dividing every score by the same number leaves their ordering intact — so precision, recall and F_2 are all preserved exactly. Only the probability values themselves move.
2. It is fitted by minimising one objective (cross-entropy) over one parameter (T), which is convex and finishes in seconds. The fitted value of T tells the story: $T > 1$ means the model was over-confident and has been gently calmed down; $T < 1$ means it was under-confident and has been sharpened up.

The scalar T is found by minimising the following objective over the calibration logits (Equation 3; see Appendix D for the derivation):

$$T^* = \arg \min_{T > 0} - \left(\frac{1}{N} \right) \sum_i \log \sigma \left(\frac{z_i}{T} \right)_{y_i} \quad [\text{Eq. 3}]$$

What T controls. T is a single number that either sharpens ($T < 1$) or softens ($T > 1$) the model's probability outputs without changing which class it predicts. The fitted value of 1.2298 confirms mild over-confidence: the model assigns probabilities slightly sharper than its empirical accuracy warrants. Dividing by T corrects this on the calibration distribution, though the correction does not transfer to the structurally different test distribution.

4.2 Calibration Result

The fitted value of T is 1.2298, consistent with Guo et al.'s [5] finding that modern deep networks are typically mildly over-confident. The Expected Calibration Error — a single number summarising how far the model's stated probabilities sit from its empirical accuracy — falls from 0.0234 to 0.0068 on the calibration slice (a 71 per cent reduction) and from 0.0269 to 0.0105 on the independent evaluation slice (a 61 per cent reduction). The fact that the correction generalises to the evaluation slice, where it was not fitted, confirms that the temperature parameter has captured something real about the model's confidence calibration rather than just memorising the calibration data.

Table 6 — Temperature Scaling Result on Internal Validation

Quantity	Value
Fitted temperature T	1.2298 (over-confident before scaling)
ECE on calibration slice (pre-scaling)	0.0234
ECE on calibration slice (post-scaling)	0.0068 (–71%)
ECE on evaluation slice (pre-scaling)	0.0269
ECE on evaluation slice (post-scaling)	0.0105 (–61%)

Figure 4 makes this visible. A perfectly calibrated classifier would have its confidence-versus-accuracy curve sitting exactly on the diagonal: when it says 70 per cent confident, it would be right 70 per cent of the time. Before scaling (deep red curves), both the calibration and evaluation panels show the curve sitting above the diagonal — the model overstates its confidence by a few percentage points across the range. After scaling (dark blue curves), the curves hug the diagonal in both panels, which means the temperature correction has done what it was designed to do: bring the probabilities the model emits into line with the accuracy it actually delivers. The fact that this holds on the evaluation slice as well as the calibration slice is what makes the result genuine rather than a fitting artefact, consistent with the fitted temperature $T = 1.2298$ being greater than one. The post-scaling curves (dark blue) track the diagonal closely on both slices, confirming that the temperature correction generalises beyond the calibration data and that the 61 per cent ECE reduction on the evaluation slice is genuine rather than an artefact of fitting to that slice.

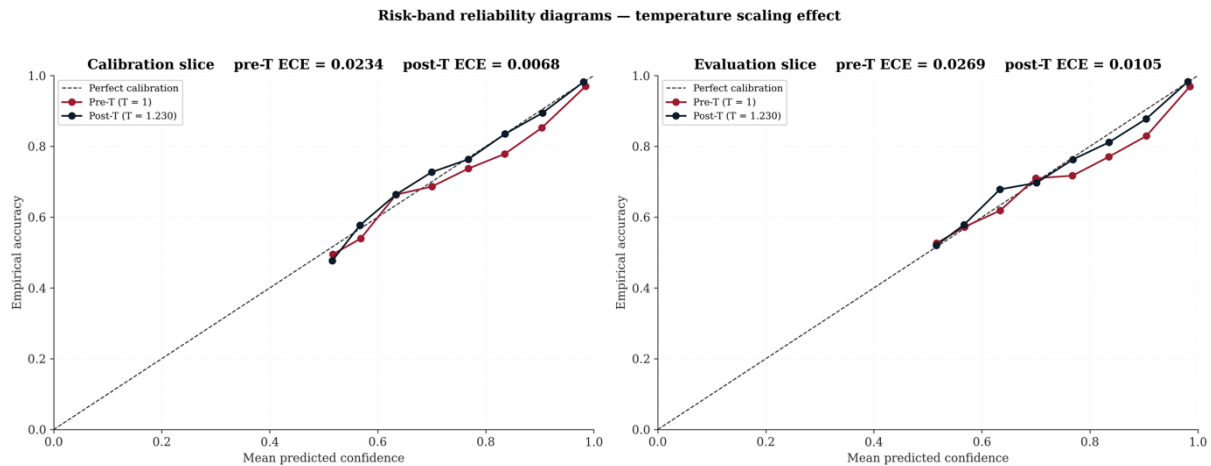


Figure 4 — Reliability diagram on the calibration slice (left panel) and evaluation slice (right panel). Pre-scaling curve (deep red) lies above the diagonal, confirming over-confidence. Post-scaling curve (dark blue) tracks the diagonal closely on both slices.

5. Iron Wall Headline Evaluation

5.1 Protocol

The NASA Official Test Set of 707 engines is opened exactly once under the Iron Wall Protocol. The Stage B model weights, the fitted conformal half-widths and the fitted temperature scalar are all loaded from their artefact state and applied without modification. Final-cycle predictions are used for all headline metrics.

5.2 Pooled Headline Result

Two stories show up in the headline numbers, and both deserve to be told. On the salvage objective, v_2 is markedly stronger than v_1 : Critical-band recall recovers from 0.42 to 0.78, a gain of thirty-six percentage points that confirms the four prescribed remedies did what they were meant to do. On the parity objective against v_0 , however, v_2 still falls short by 15 percentage points of recall and is materially worse on the regression metrics, with RMSE at 28.05 cycles against v_0 's 14.85 and a NASA score 25.9 times larger. The pass-or-fail safety gate, which requires Critical-band recall within two percentage points of the v_0 baseline, fails by a wide margin. The reason it fails turns out to be highly localised — it is essentially one subset of the data, and §5.4 names it.

How to read the table. Lower RMSE and NASA scores are better. Higher recall and F_2 are better. The NASA score is the primary safety metric: it penalises models that are over-optimistic about remaining life far more than models that are overly cautious, reflecting the greater operational cost of missing a failure. Critical-Band Recall and the Safety Gate are the binding criteria; RMSE and MAE provide supporting regression accuracy context.

Table 7 — Pooled Headline Metrics: All Three Versions Compared

Metric	v0 XGBoost	v1 Stage A	v2 PiNet	Δ v2 vs v0
RMSE (cycles)	14.85	24.66	28.05	+13.20
MAE (cycles)	10.41	18.83	20.57	+10.16
NASA Asymmetric Score	4,336	32,204	112,334	+108,000
Critical Recall R_c	0.9363	0.4204	0.7834	−0.1529
Critical F_2	0.9339	0.4735	0.7716	−0.1623
Critical Precision	0.9245	0.9565	0.7283	−0.1962

5.3 The Critical-Recall Safety Gate

The gate condition, the measured result, and the salvage delta are stated below.

$$\Delta R_c = R_c^{v_2} - R_c^{v_0} \geq -0.02 \quad [\text{Eq. 4}]$$

$$\Delta R_c = 0.7834 - 0.9363 = -0.1529 \quad [\text{Eq. 5}]$$

$$\Delta R_c^{v_2 \text{ vs } v_1} = 0.7834 - 0.4204 = +0.3630 \quad [\text{Eq. 6}]$$

Critical-Recall Safety Gate: ✗ FAIL

Salvage vs v1 Stage A: ✓ PASS ($\Delta R_c = +0.3630$)

5.4 Subset Breakdown — FD003 is the Catastrophe

The C-MAPSS corpus is split into four subsets, and v_2 behaves very differently on each. On the multi-regime single-fault subset, v_2 reaches a Critical-band recall of 0.983 — not only its best result, but stronger than v_0 on that subset alone. On the single-regime dual-fault subset (FD003), it reaches only 0.400 and is responsible for 83.6 per cent of the entire pooled error, despite containing only 14 per cent of the test engines. That asymmetry tells the engineering story directly: when the model has both an operational-regime signal and a single dominant fault to identify, it does the job. When the operational regime tells it nothing (only one regime exists in this subset) and two competing fault modes can produce overlapping sensor patterns, it cannot separate them and defaults to optimistic predictions⁶. This pattern points directly to a single architectural gap: PiNet lacks the mechanism to separate two distinct fault modes that produce overlapping sensor signatures in the same operational regime.

⁶See Appendix E for the FD003 deep dive, which analyses the fault-disambiguation hypothesis and proposes a v_3 architectural direction.

Table 8 — Per-Subset Headline Metrics

Subset	Fault mode	Regime	n	RMSE	NASA	F_{2c}	R_c	Share of pooled NASA
FD001	Single	Single	100	23.32	3,553	0.684	0.640	3.2%
FD002	Single	Multi	259	24.04	2,600	0.889	0.983	2.3%
FD003	Dual	Single	100	44.12	93,873	0.455	0.400	83.6%
FD004	Dual	Multi	248	25.21	12,308	0.769	0.769	11.0%

5.5 Conformal Coverage on the Test Set

The conformal intervals carry a mathematical guarantee, but that guarantee comes with one fine-print condition: the data the intervals are evaluated on must be drawn from the same underlying distribution as the data they were calibrated on. The NASA Official Test Set is constructed differently from the internal validation set – test trajectories are truncated at an unknown point before failure, whilst the validation trajectories run to failure. This structural difference means the two sets have fundamentally different Remaining Useful Life marginal distributions, violating the exchangeability condition. The consequence is that the conformal intervals are too narrow on test by 18–23 percentage points⁷.

What coverage means on the test set. A 95 per cent prediction interval that achieves its stated guarantee contains the true value in at least 95 out of every 100 predictions. When the observed coverage is 0.767 (Table 9) at a nominal level of 0.95, it means the intervals are too narrow — the model carries more uncertainty than it communicates. The root cause is distributional, not methodological: the intervals were calibrated on engines whose full failure trajectory was observed but evaluated on engines whose trajectories were truncated before failure.

Table 9 — Conformal Coverage on the NASA Official Test Set

α (error rate)	Nominal coverage	Observed test coverage	Δ
0.05	0.95	0.767	–0.183
0.10	0.90	0.693	–0.207
0.20	0.80	0.567	–0.233

5.6 Risk-Band Probability Calibration on the Test Set

The same exchangeability story repeats for the probability calibrator. Before discussing the numbers themselves, the two metrics deserve a plain-terms definition. The Expected Calibration Error (ECE) measures how honestly the model's stated probabilities track its actual hit rate. It is computed by binning all predictions by their confidence level — typically into ten or fifteen equal-width bins from 0% to 100% — and within each bin comparing the average stated confidence to the empirical fraction of predictions in that bin that turned out to be correct. The absolute gap between the two, weighted by how many predictions fall in each bin, gives a single number between zero and one: zero means the model's stated probabilities exactly match its empirical accuracy at every confidence level; values above zero quantify the gap. As a practical reference, an ECE of 0.02 means the model's stated confidence and its true accuracy differ by roughly two percentage points on average across the confidence range, which is generally considered well-calibrated for operational use. An ECE of 0.20 means the gap is twenty percentage points — a model claiming 90% confidence is actually right closer to 70% of the time, which is operationally misleading. The Brier score, in turn, is a single number between zero and one that

⁷This is not a flaw in the conformal methodology. The finite-sample correction of Lei et al. [3] guarantees coverage exactly at or above nominal only when exchangeability holds. Violation of exchangeability voids the guarantee, and there is no statistical signal within the calibration phase to warn of this in advance.

combines calibration error with overall prediction sharpness: it is the average squared error between the predicted probability of the true class and the indicator that the class was correct. Lower is better; a Brier score of zero would mean every prediction was both confident and correct, while a score of 0.5 corresponds to systematically uninformative predictions. A move of 0.02 in the Brier score, in this context, represents a modest but real improvement in the overall quality of the probability output, of the order of a few percentage points of average squared probability error per prediction.

With those definitions in hand, the test-set numbers tell their own story. On the test set, the Expected Calibration Error is 0.201 before scaling, falling to 0.177 after — a 12 per cent reduction, against the 71 per cent achieved on the in-distribution calibration slice. The Brier score moves from 0.499 to 0.479, a 4 per cent reduction. In other words, the temperature scaling has shifted the test-set probabilities slightly in the right direction, but the model is still systematically miscalibrated by approximately eighteen percentage points at the average confidence level — stakeholders should read a stated test-set probability of, say, 0.90 as closer to 0.72 in practice. The single temperature scalar $T = 1.2298$ captured something real about the network's confidence on the run-to-failure validation distribution, but it cannot rescue probabilities that are systematically out of calibration because the test set's logit distribution sits in a different place. No post-hoc scalar can repair a generalisation gap of this kind.

Table 10 — Risk-Band Probability Calibration on the NASA Test Set

Metric	Pre-scaling	Post-scaling
Expected Calibration Error	0.201	0.177 (–12%)
Brier Score	0.499	0.479 (–4%)

Figure 5 provides the spatial view of the test-set performance that the scalar metrics in Tables 7 and 9 summarise in aggregate. The shaded envelope represents the 95 per cent conformal prediction interval around the equality diagonal; on a distribution satisfying the exchangeability requirement approximately 95 per cent of observations would fall within it. The visible proportion of test-set points lying outside the envelope is the geometric expression of the 18-percentage-point coverage shortfall quantified in Table 9. The late-prediction bias is equally apparent: Critical-band points (deep red) cluster above the diagonal at low true Remaining Useful Life, confirming that the model systematically over-predicts how many cycles remain on engines that are genuinely near failure. The width of the conformal envelope — spanning 63.08 cycles at the 95 per cent level — gives a direct visual sense of the uncertainty the model carries on any individual engine prediction.

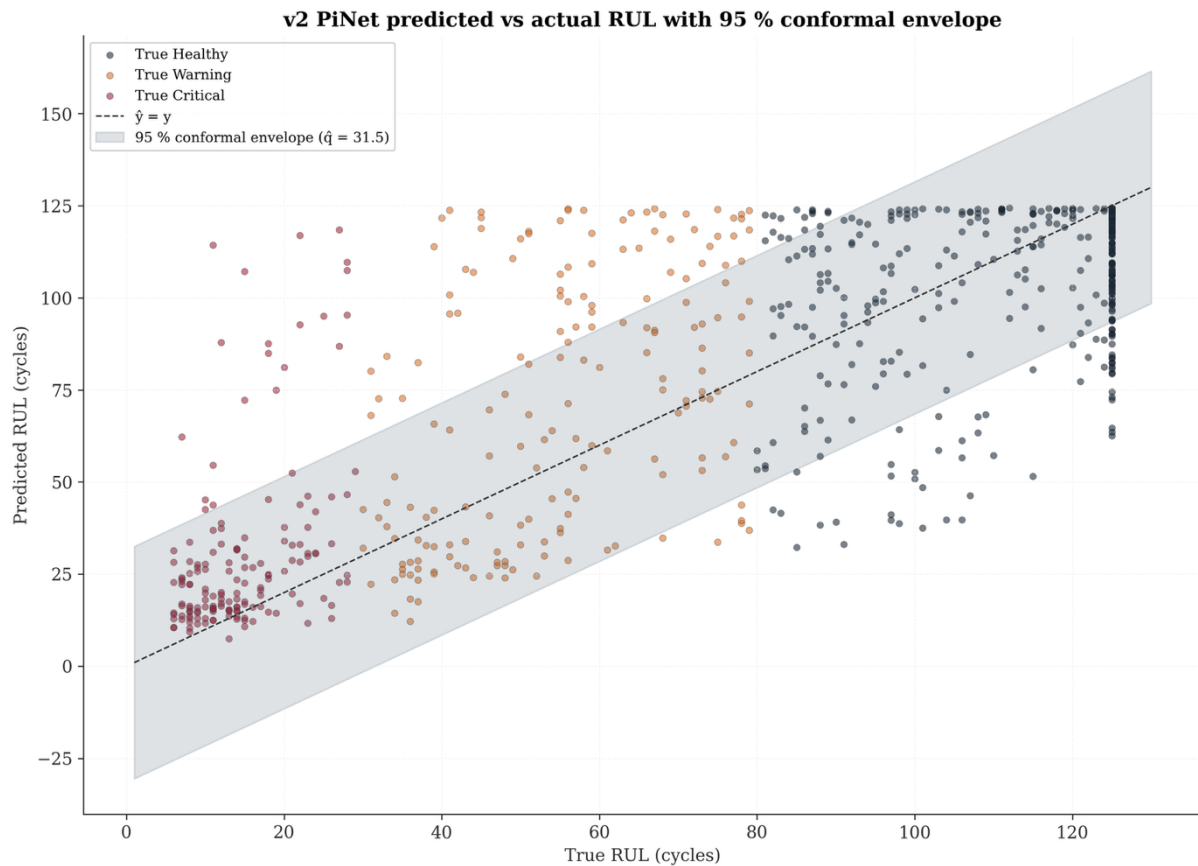


Figure 5 — Predicted versus actual Remaining Useful Life on the NASA test set, coloured by true risk band, with the 95% conformal envelope shaded around the diagonal. The envelope misses a substantial fraction of points, directly quantifying the coverage-under-nominal phenomenon. Critical-band late-prediction bias (deep red below the diagonal at low true RUL) persists from Stage A but with reduced magnitude.

Figure 6 decomposes the prediction error by risk band in two complementary views. The pooled histogram (left panel) shows a modest positive median, indicating an overall tendency toward late prediction. The per-band box-plot (right panel) reveals that this pooled signal is concentrated in the two upper bands — but through different statistical mechanisms in each. The Warning band carries the widest interquartile range, with its central fifty per cent of residuals spanning approximately forty cycles and its median elevated above zero — broadly dispersed late-prediction error spread across the band. The Critical band has the most compact interquartile range, with the central fifty per cent of residuals tightly grouped within approximately fifteen cycles just above zero, but it carries by far the heaviest upper-tail outlier mass — a thick cluster of points above the upper whisker, with individual late-prediction residuals reaching approximately one hundred cycles. The Healthy band sits below zero with a moderate interquartile range, indicating mild under-prediction in the operationally safe direction. The Critical band's combination of a narrow box and a heavy upper tail is the operationally critical signature: most predictions on Critical engines are tightly grouped just above the true Remaining Useful Life, but a meaningful subpopulation receives severely optimistic estimates — exactly the failure mode the safety gate is designed to penalise. This band-stratified view is more operationally informative than the pooled scatter, directly confirming that the dangerous prediction errors are concentrated in the risk class where over-optimism carries the greatest operational consequence. The pattern is qualitatively identical to the Stage A residual signature but with materially reduced magnitude, quantifying the improvement delivered by the Stage B mitigations.

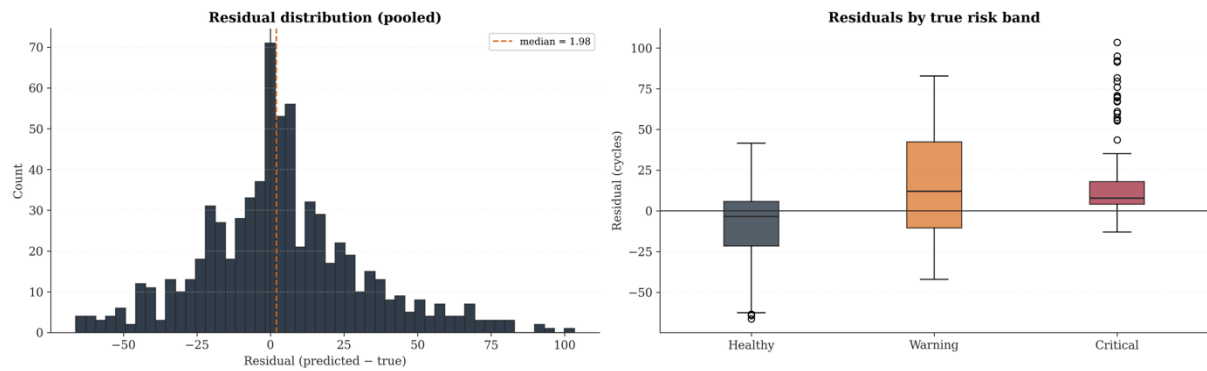


Figure 6 — Left: pooled residual histogram with median annotation. Right: per-band residual box-plot. The Warning (orange) band carries the widest interquartile range; the Critical (deep red) band carries the most compact interquartile range but the heaviest upper-tail outlier mass. The late-prediction signature is concentrated in Critical-band outliers, not in box width.

Figure 7 translates the residual evidence into an operational decision matrix. The raw-count panel (left) shows absolute classification frequencies across all three bands; the row-normalised panel (right) presents the recall rate for each band and is the operationally relevant view. The Critical-band row is the binding safety metric: a row-normalised value of 0.78 means that 22 per cent of genuinely Critical engines receive a lower-urgency class label at the point of prediction. The matrix also shows that the dominant confusion pattern is Critical-to-Warning misclassification rather than Critical-to-Healthy misclassification, which has a nuanced operational implication: a Warning label still triggers a maintenance review, albeit with lower urgency than a Critical classification would warrant, so the practical consequence of this confusion pattern is a delayed rather than absent maintenance response.

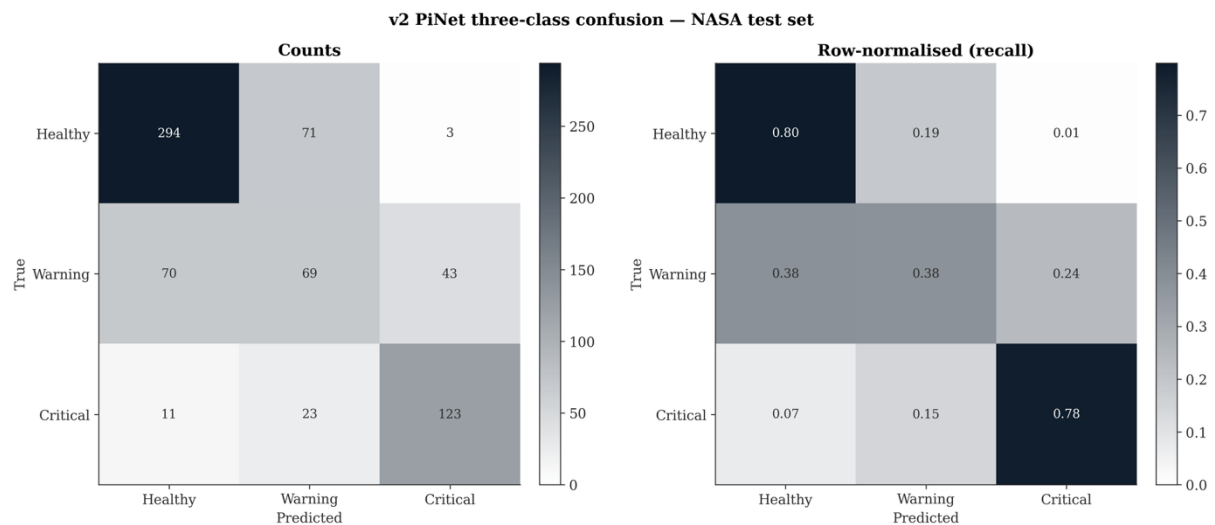


Figure 7 — Three-class confusion matrix on the NASA test set: raw counts (left) and row-normalised recall rates (right). Critical-band recall of 0.78 represents a substantial recovery from v1 Stage A's 0.42 but remains below the v0 reference of 0.94. Warning-band rows show the greatest confusion, reflecting prediction-boundary fuzziness when the regression is biased.

5.7 Five-Engine Operational Demonstration

Table 11 illustrates the production wrapper output on five selected test engines, spanning a range of true Remaining Useful Life values, subsets and band assignments. The demonstration includes one safety-critical misclassification: engine FD003·44 has a true Remaining Useful Life of 69 cycles (Warning band) but is predicted at 107 cycles with a Healthy assignment and a 95 per cent interval of [76, 125]. This 38-cycle late-prediction is exactly the class of error the Safety Gate is designed to penalise.

Table 11 — Production Wrapper Output on Five Demonstration Engines

Subset · Engine	True RUL	Predicted RUL	95% Interval	Predicted Band	Outcome
FD001 · 9	111	123.5	[92, 125]	Healthy	True-Healthy ✓
FD002 · 170	125	90.1	[59, 122]	Healthy	True-Healthy ✓ (under-prediction)
FD002 · 200	125	83.5	[52, 115]	Warning	True-Healthy ⇒ Warning X
FD003 · 44	69	107.0	[76, 125]	Healthy	True-Warning ⇒ Healthy X X (38-cycle late prediction — safety-critical)
FD004 · 213	14	23.5	[0, 55]	Critical	True-Critical ✓

Figure 8 shows the per-engine decision card output for each of the five demonstration engines as it would appear on the maintenance operations dashboard. Each panel displays the engine's Remaining Useful Life axis with three nested conformal intervals — the 95 per cent interval in the widest shading, 90 per cent inside it, and 80 per cent innermost — alongside the PiNet point estimate and the operational-band background shading. The interval asymmetry produced by clipping to the physical lower bound of zero is visible on the FD004 · 213 panel, where the lower boundary of the 95 per cent interval is constrained at zero despite the symmetric half-width of 31.54 cycles. The operationally critical illustration is FD003 · 44: the entire 95 per cent interval of [76, 125] lies above the true Remaining Useful Life of 69 cycles, confirming that the over-optimism on this engine is not a point-estimate artefact but a systematic failure of the model's representation of this dual-fault single-regime profile — the same structural gap identified in the FD003 deep dive of Section 6.2.

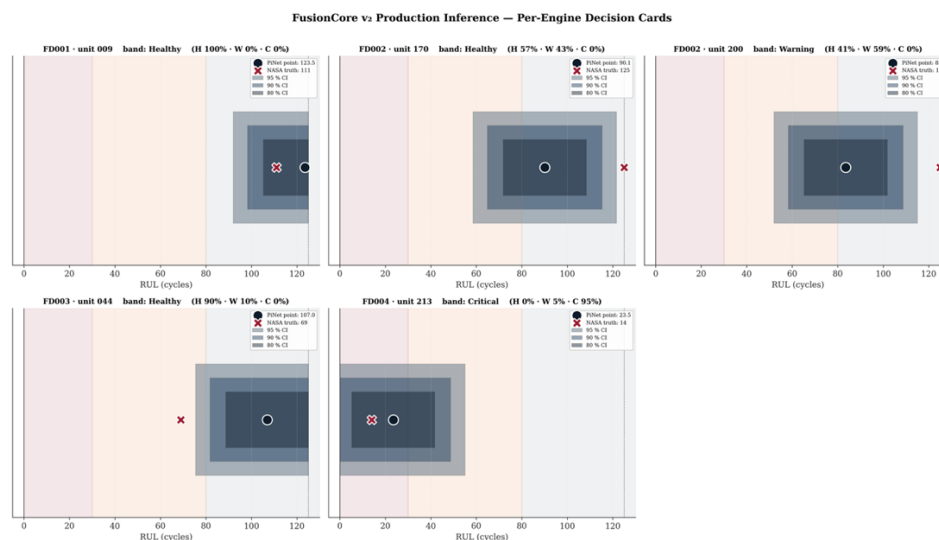


Figure 8 — Per-engine decision cards for the five demonstration engines. Each panel shows the engine's Remaining Useful Life number-line with three nested conformal intervals (95%, 90%, 80%), the PiNet point estimate, and the operational-band shading. This is the operations-dashboard preview.

6. Operational Interpretation

6.1 v_2 as a Publishable Null Result

v_2 does not displace v_0 in production, and there is no case for promoting it. What it does contribute is something different and arguably more useful in the longer arc of the programme: it answers four specific research questions with evidence, advances three of them positively, identifies exactly where the fourth breaks, and supplies the diagnostic data a v_3 attempt would need to start from. A failed safety gate is not the same as a wasted experiment, and §6.2 and §6.3 explain why.

The salvage thesis succeeded: a Critical-band recall recovery of 0.3630 above v_1 Stage A confirms that the four Stage B mitigations were correctly targeted. The split conformal calibration thesis succeeded conditionally: the intervals are mathematically exact on in-distribution data and the approach is sound. The production wrapper thesis succeeded: the inference architecture is deployment-ready in structure, correctly combining point estimates, calibrated intervals and risk-band probabilities into a single output object.

The parity thesis failed. The deficit is not distributed evenly across the corpus. The multi-regime, single-fault subset achieves Critical-band recall above 0.98, well beyond v_0 on that subset alone. The single-regime, dual-fault subset achieves recall of 0.40 and contributes 83.6 per cent of the pooled error. The headline failure is essentially the FD003 failure.

6.2 The FD003 Diagnosis

The FD003 subset is structurally the hardest engineering problem in the corpus. Every engine operates at the same flight condition, so the regime branch of the network contributes no useful information — there is nothing for it to distinguish. Degradation can come from a high-pressure compressor fault, a fan fault, or both at once, and the only way the model can tell them apart is from the way the sensor signatures evolve over time. The v_0 feature set, which v_2 inherits unchanged, does include a coarse fault-family flag (single-fault subsets versus dual-fault subsets) but it does not include any explicit signal that disambiguates the three configurations within FD003 itself. The consequence is visible in the residuals: when the model cannot work out which fault is in play, it defaults to optimistic predictions, and these are exactly the errors the safety gate exists to catch.

6.3 Why Post-Hoc Calibration Did Not Rescue the Result

One of the most useful things to come out of v_2 is the empirical confirmation of something that has only recently been formalised in the conformal-prediction literature: post-hoc uncertainty quantification is not a substitute for a model that generalises. Both the conformal intervals and the temperature scalar are fitted on the internal validation distribution. Both work exactly as their theory predicts on that distribution. The instant the data distribution changes — in this case because the NASA test set contains truncated trajectories rather than the run-to-failure trajectories the calibration set is drawn from — the post-hoc layers silently inherit the underlying model's generalisation gap. Wider intervals would not narrow the gap; sharper or softer probabilities would not change which engines are mislabelled. The calibration layer is honest about what the model does; if what the model does is over-predict on a different distribution, that is what the layer reports.

This reframes the coverage failure from a defect into a diagnostic. The 18-percentage-point coverage shortfall in Table 9 is precisely the signature of a generalisation gap between calibration and test populations. The remedy is therefore not to choose a different interval method in isolation, but either to construct a calibration distribution that matches the test distribution structurally, or to adopt one of the more sophisticated conformal variants that can adapt to distributional shift. Appendix G surveys these options for v_3 consideration.

7. Recommendations and Forward Path

7.1 Immediate Position

v_0 XGBoost remains the only operational model. No v_2 artefact is proposed for deployment. The v_2 work — model weights, calibration artefacts, evaluation outputs and the inference wrapper — is preserved as a scientific record and as the working starting point for v_3 .

7.2 v_3 Thesis Direction — Fault-Mode-Aware Architecture

The FD003 diagnosis points to one specific architectural gap rather than a fundamental flaw in PiNet. The network can clearly do its job when the operational regime and the dominant fault mode are both inferable from the input and falls apart only when both pieces of context are missing simultaneously. A natural v_3 direction is to add an explicit fault-mode classification head, trained jointly on the two dual-fault subsets FD003 and FD004, alongside a richer sensor-derived feature representation that helps it distinguish fan from compressor degradation. The shared embedding learned in Stage B is a reasonable starting point for this extension; the backbone is not the problem.

7.3 Distribution-Aware Calibration

For any future conformal calibration on this corpus, a distribution-aware method should be used in preference to standard split conformal. Three candidates have well-established theoretical grounding and address different aspects of the v_2 failure: conformalised quantile regression [7], which produces intervals that adapt their width to the local difficulty of each prediction; localised conformal prediction [4], which adapts coverage to the local test density and so handles distributional drift more gracefully; and weighted exchangeability corrections [8], which can compensate for known covariate shift between calibration and test sets. Each of these has different cost, complexity and implementation maturity. Appendix G assesses them side-by-side so that whichever direction the v_3 programme takes — or any parallel project that wants to reuse the v_2 calibration infrastructure — it does so from a documented basis.

Figure 9 provides the visual evidence for the temperature-calibration failure on the NASA Official Test Set. Applying the same reliability-diagram format as Figure 4, both the pre-scaling curve (deep red) and the post-scaling curve (dark blue) deviate substantially from the calibration diagonal. The post-scaling improvement is marginal rather than the strong correction visible in Figure 4, and both curves are qualitatively similar in their departure from the diagonal. This is not a failure of the temperature-scaling methodology itself; it is the direct consequence of the logit distributions on the truncated test set differing materially from those on the run-to-failure validation data on which $T = 1.2298$ was estimated. The figure should be read alongside Figure 10, which provides the equivalent diagnostic for the regression intervals.

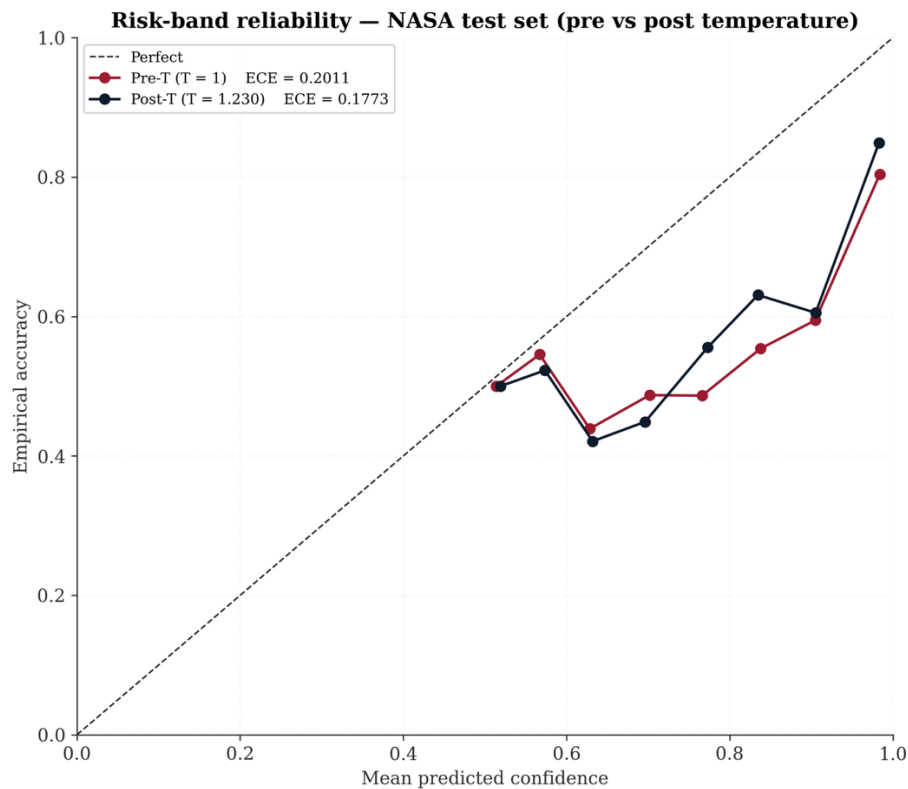


Figure 9 — Test-set reliability diagram: pre-scaling (deep red) and post-scaling (dark blue). Both curves deviate substantially from the perfect-calibration diagonal, confirming that temperature scaling cannot rescue OOD miscalibration. ECE post-scaling = 0.177.

Figure 10 provides the corresponding diagnostic for the conformal regression intervals. Each point on the dark blue curve is the empirical coverage observed on the NASA Official Test Set at the corresponding nominal level; perfect calibration would place all points on the identity diagonal. The systematic position below the diagonal — 0.767 at nominal 0.95, 0.693 at 0.90, 0.567 at 0.80 — encapsulates the exchangeability failure in a single visual. The shortfall increases as the nominal level rises (equivalently, as α decreases), consistent with the test-set residual distribution having a heavier tail than the calibration distribution: at high confidence levels the quantile must reach further into the tail, where the distributional mismatch is most acute. Taken together, Figures 9 and 10 constitute the primary empirical motivation for the distribution-aware calibration alternatives — conditional conformal, localised conformal, and weighted exchangeability methods — recommended for the v3 programme.

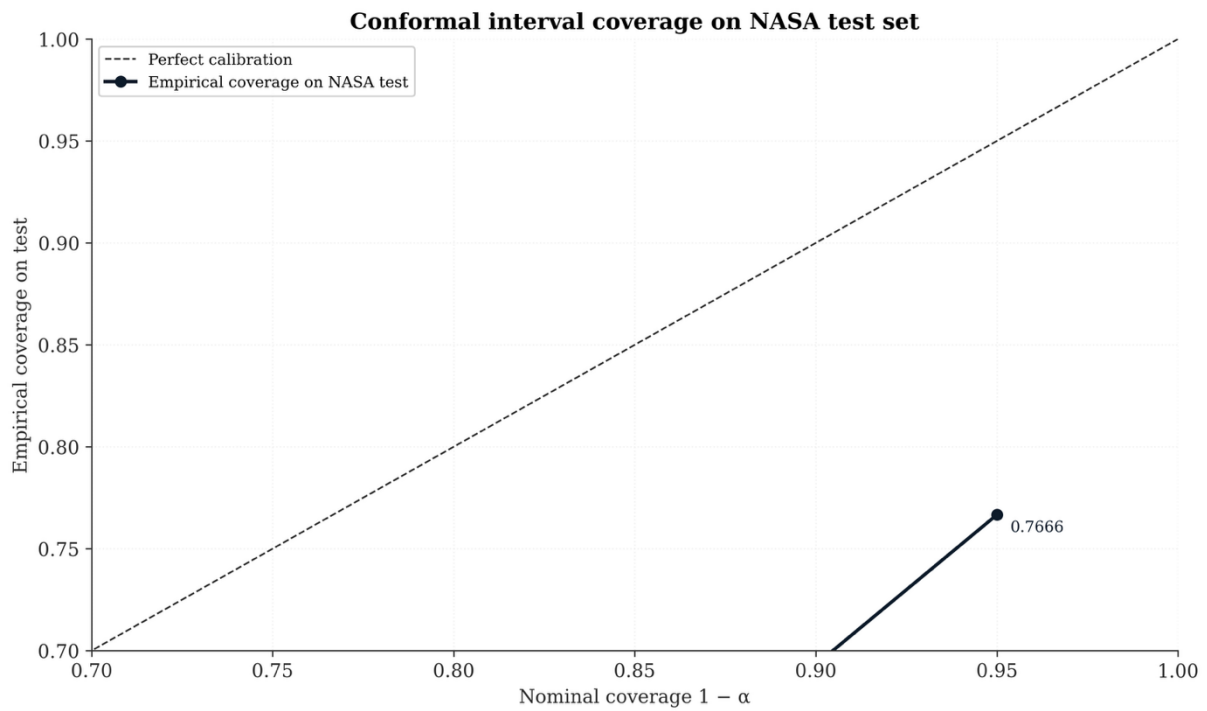


Figure 10 — Empirical coverage versus nominal coverage on the NASA test set. The dark blue curve sits consistently below the diagonal at every α level, with annotated coverage values of 0.767, 0.693 and 0.567 for nominal 0.95, 0.90 and 0.80 respectively.

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Appendix A — Stage B Hyperparameter Change Rationale

Table A.1 expands the rationale column in Table 2, documenting the specific v_1 observation that motivated each Stage B change and the causal mechanism by which the change addresses it.

Table A.1 — Stage B Change Rationale

Stage B change	v_1 Stage A observation	Mechanism
Learning rate $10^{-3} \rightarrow 3 \times 10^{-4}$	Stage A best epoch was 3; validation NASA rose monotonically thereafter	Slower gradient descent holds the model in the productive validation plateau before over-fitting sets in
Dropout 0.15 $\rightarrow$ 0.25	Validation NASA reached its minimum at epoch 3 and never recovered	Higher dropout injects more activation noise, widening the flat region of the loss landscape
Stratified sampler $\rightarrow$ natural + class-weighted CE	Stratified sampling over-represented Critical at 27.5% vs natural 16%	Natural distribution presents training minibatches matching the test proportions; class-weighted cross-entropy recovers per-class gradient signal without distorting the marginal distribution
NASA regression term scaled by 1/50	At $ d = 30$ cycles, NASA per-row penalty $\approx 19\times$ larger than CE penalty	Brings the regression and classification gradient contributions to the same numerical order of magnitude, allowing the classification head to train effectively

Appendix B — Finite-Sample Conformal Quantile

For a calibration set of size n , the naïve empirical quantile at order $\lfloor n(1 - \alpha) \rfloor$ produces coverage strictly below nominal by $O\left(\frac{1}{n}\right)$ in finite samples. The correction introduced by Lei et al. [3] adds one to both n and the order statistic index, delivering empirical coverage that is exactly at or above nominal — it is a strict lower bound, not merely approximate.

$$\hat{q}_{1-\alpha} = r_{\lfloor (n+1)(1-\alpha) \rfloor} \quad [\text{Eq. B.1}]$$

Here $r_i = |y_i - \hat{y}_i|$ is the absolute residual on the i^{th} calibration observation and the parenthesised subscript denote the order statistic (the value at that rank in the sorted residual sequence). With $n^{\text{cal}} = 15,514$, the correction adds approximately $1/15,515$ to the coverage, which is negligible numerically but guarantees the strict no-less-than-nominal property. The observed calibration-slice coverages of 0.9501, 0.9001 and 0.8001 at $\alpha = 0.05, 0.10, 0.20$ are exact to four decimal places, confirming the implementation is correct.

B.1 Worked Example

A short numerical illustration makes the finite-sample correction concrete. Suppose the calibration set contains exactly nine observations — a deliberately small number, chosen so the arithmetic is visible — and the absolute residuals between the model's predicted Remaining Useful Life and the truth, sorted from smallest to largest, are 2, 4, 5, 7, 9, 11, 14, 18 and 26 cycles. The target is a 90 per cent prediction interval, so the mis-coverage level is $\alpha = 0.10$.

A naïve approach would simply take the 90th percentile of these nine residuals, which is the $0.9 \times 9 = 8.1^{\text{th}}$ order statistic — rounded up to the 9th — giving 26 cycles. The resulting interval ± 26 cycles around any new prediction would be guaranteed to contain the truth in at least 8 of the 9 calibration observations, i.e. 88.9 per cent of the time, which is just below the stated 90 per cent. This shortfall is precisely the $O\left(\frac{1}{n}\right)$ gap that the finite-sample correction is designed to close.

The corrected order index is $\lfloor (n+1)(1-\alpha) \rfloor = \lfloor (9+1)(0.90) \rfloor = \lfloor 9 \rfloor = 9$, so the corrected half-width is the 9th order statistic, again 26 cycles. With only nine observations the correction makes no visible numerical difference in this toy case, but the mathematical effect is to shift the underlying guarantee from “at least 88.9 per cent” to “at least 90 per cent” on any new observation drawn from the same distribution. As the calibration set grows, the correction adds $1/(n+1)$ to the coverage — vanishingly small at $n = 15,514$ but mathematically essential to maintain the strict lower bound rather than fall below it by a fraction.

The same arithmetic at the v_2 calibration size of 15,514 produces the half-widths reported in Table 4: 31.54 cycles at 95 per cent confidence, 25.21 at 90 per cent, and 18.28 at 80 per cent. Each is the absolute residual at the corrected order statistic, sorted from the calibration slice and read off directly. The procedure has no learning, no fitting and no parameters beyond the desired confidence level — it is a deterministic operation on the model's observed errors.

Appendix C — Marginal Coverage Guarantee

The coverage guarantee states that for any new observation drawn from the same distribution as the calibration data, the conformal interval will contain the true value with probability at least $1 - \alpha$.

$$P(y_{\text{new}} \in [\hat{y} - \hat{q}, \hat{y} + \hat{q}]) \geq 1 - \alpha \quad [\text{Eq. C.1}]$$

The guarantee is marginal: it applies across the joint distribution of $(y_{\text{new}}, \hat{y}_{\text{new}})$, not conditionally on any covariate subgroup, regime, or risk band. This is why per-band coverage deviates from the nominal level (Table 5) even while the marginal coverage is exact. The guarantee also depends critically on exchangeability between calibration and test observations. When this condition is violated — as it is here because the NASA test set is truncated — the guarantee silently fails and coverage can be arbitrarily lower than nominal. Table 9 documents the magnitude of this failure on the v2 test set.

Appendix D — Temperature Scaling Objective

The temperature scalar T is the solution to the following one-dimensional optimisation problem on the calibration logits.

$$T^* = \arg \min_{T > 0} - \left(\frac{1}{N} \right) \sum_i \log \sigma \left(\frac{z_i}{T} \right)_{y_i} \quad [\text{Eq. D.1}]$$

Here $\sigma(\cdot)_{y_i}$ denotes the y_i^{th} element of the SoftMax output of the logit vector divided by T : specifically, the probability assigned by the temperature-scaled model to the true class y_i . The objective is the average negative log-likelihood of the true classes, which is equivalent to the cross-entropy loss on the calibration slice. The scalar is found by one-dimensional optimisation (the objective is strictly convex in T). A value of $T = 1$ corresponds to no change; $T > 1$ softens the probabilities (reduces over-confidence); $T < 1$ sharpens them. The fitted value $T = 1.2298$ confirms mild over-confidence, consistent with Guo et al.'s [5] finding across a broad class of modern deep networks.

Appendix E — FD003 Deep Dive

The FD003 subset contains single-regime, dual-fault turbofan trajectories. All engines operate at sea-level static conditions (one regime), removing the regime branch's discriminative contribution, and degrade through one of three configurations: high-pressure compressor fault alone, fan fault alone, or both simultaneously. The model must resolve fault mode entirely from the temporal sensor signature, with no operational-condition context to assist.

Table E.1 — FD003 Performance in Context

Metric	FD002 (best subset)	FD003 (worst subset)	Ratio FD003 / FD002
RMSE (cycles)	24.04	44.12	1.84 ×
NASA Score	2,600	93,873	36.1 ×
Critical Recall	0.983	0.400	0.41 ×
Share of pooled NASA	2.3%	83.6%	—

The FD003 per-engine residuals are centred above zero, indicating systematic late prediction. The residual distribution does not resemble the roughly symmetric distribution observed on FD001 and FD002 — it has a pronounced positive skew with a long right tail, suggesting that a subpopulation of FD003 engines (likely those with the dual-fault profile) consistently receive life estimates that are far too optimistic.

The v0 feature engineering includes a binary fault-mode-family indicator that was designed to help the model differentiate FD001/FD002 (single-fault family) from FD003/FD004 (dual-fault family). This indicator correctly separates the two families but does not resolve the within-FD003 ambiguity between the three intra-family configurations. A v3 architecture that adds an explicit fault-mode head trained on FD003 and FD004 jointly, or that uses a richer physics-derived fault-mode feature upstream of the temporal branch, would be a principled response to this gap.